

Premier University
Department of Computer Science and Engineering
8th Semester (B.Sc.) Final Examination, Spring 2025

Course Code: CSE 453

Course Title: Compiler Construction

Time: 3 Hours

Full Marks: 40

N.B.: Question 1 is compulsory. And answer any three (3) questions from the remaining ones. Each question carries equal marks.

- Q1. a.** Discuss the common types of errors encountered during syntax analysis with CO4 4
 examples. How does a parser recover from syntax errors?
- b.** Does left factoring make a grammar suitable for top-down parsing? Explain the CO4 6
 process of left factoring a grammar and left factor the following grammar:

$$S \rightarrow aAd \mid aB$$

$$A \rightarrow a \mid ab$$

$$B \rightarrow ccd \mid ddc$$

- Q2. a.** Consider the following grammar and associated parsing Table: 2(a) CO3 5

State	Action				Goto	
	a	b	c	\$	S	B
0. $S' \rightarrow S$			s2		1	
1. $S \rightarrow SaB$	s3			acc		
2. $S \rightarrow c$	r2			r2		
3. $B \rightarrow b$		s4				7
4. $B \rightarrow bB$	r3	s6		r3		5
	s5			r4		
	r3			r3		
	r1			r1		

Table: 2(a)

Show how the string **cabbab\$** is parsed. [Hints: show the contents of the stack, input, and corresponding actions in a table.]

- b.** Compute the canonical set of LR(0) items for the following grammar: CO2 5

$$S \rightarrow (L) \mid id$$

$$L \rightarrow L, S \mid S$$

- Q3. a.** Eliminate the left recursion from the following grammar: CO2 5

$$S \rightarrow Aa \mid b$$

$$A \rightarrow Ac \mid Sd \mid \epsilon$$

Then, compute the FIRST and FOLLOW sets for all non-terminals.

- b. Describe the process of constructing an annotated parse tree and its significance in CO3
syntax-directed translation. Draw an annotated parse tree for $5 * 7 * 9$ considering
the following SDD as shown Fig: 3(b).

PRODUCTION	SEMANTIC RULES
1) $T \rightarrow F T'$	$T'.inh = F.val$ $T.val = T'.syn$
2) $T' \rightarrow * F T'_1$	$T'_1.inh = T'.inh \times F.val$ $T'.syn = T'_1.syn$
3) $T' \rightarrow \epsilon$	$T'.syn = T'.inh$
4) $F \rightarrow \text{digit}$	$F.val = \text{digit.lexval}$

Fig. 3(b)

- Q4. a. Compare and contrast syntax trees and parse trees. Construct a syntax tree for the CO3 5
following arithmetic expression-

$$(a + b) * (c - d) + ((e/f) * (a + b))$$

- b. Discuss the advantages of using a Directed Acyclic Graph (DAG) for representing CO3 5
expressions. Convert the following arithmetic expression into a DAG
representation:

$$a = b + c * d + b + c * d$$

Explain how DAG helps in eliminating redundancy.

- Q5. a. Consider the following statement: CO2 1.5*4
=6

```
do {
    i = i + 1;
} while (a[i] < v);
```

Translate these statements into:

(i) Three-address code, (ii) Quadruples, (iii) Triples, and (iv) Indirect triples
representation.

- b. Explain static and dynamic type checking with examples. Describe the functions of CO2 4
a type checker and its role in compiler phases.

- Q6. a. What is instruction selection? How does the level of the intermediate CO2 2
representation (IR) affect the complexity of instruction selection?

- b. Do the following from the given intermediate code that turns a 10x10 Matrix A CO2 5+3=
into an identity matrix. 8

- i. Determine the leader instructions and partition the code into basic blocks.

```

1)  i = 1
2)  j = 1
3)  t1 = 10 * i
4)  t2 = t1 + j
5)  t3 = 8 * t2
6)  t4 = t3 - 88
7)  a[t4] = 0.0
8)  j = j + 1
9)  if j <= 10 goto (3)
10) i = i + 1
11) if i <= 10 goto (2)
12) i = 1
13) t5 = i - 1
14) t6 = 88 * t5
15) a[t6] = 1.0
16) i = i + 1
17) if i <= 10 goto (13)

```

Fig. 6(b)

- ii. Construct a flow graph to represent the flow of control between the basic blocks.

Premier University
Department of Computer Science and Engineering
8th Semester (B.Sc.) Final Examination, October 2025
Session: Spring 2025

Course Code: CSE 457

Time: 3 Hours

Course Title: Machine Learning

Full Marks: 40

NB: Answer any of four (4) from following six (6) questions.

Each question carries equal marks.

- | | | | |
|------------|--|-------|-----|
| Q.1 | a. In the context of supervised learning, what distinguishes the training data from the testing data, and why is this differentiation important? | 2 | CO1 |
| | b. i. Suppose you are predicting blood pressure $y^{(1)}$ and cholesterol level $y^{(2)}$ from age, weight, and exercise hours: <ul style="list-style-type: none">● How would you set up the model if you use multiple regression?● How would it differ under multivariate regression? | 1+1+2 | CO2 |
| | ii. A dataset has 200 students with predictors (study hours, attendance, sleep hours) and outcomes (math score, programming score). If the outcomes are highly correlated, explain why multivariate regression would give more reliable estimates than multiple regression. | | |
| | c. Explain the concept of k-fold cross-validation and its advantages over simple train-test splits in case of imbalanced dataset. | 2+2 | CO4 |
| Q.2 | a. What is the primary goal of clustering algorithms in unsupervised learning, and how do they differ from classification algorithms used in supervised learning? | 2 | CO1 |
| | b. In reinforcement learning, what are the three main components of the agent's interaction with the environment? Suppose an autonomous robot learns to navigate a maze using Q-learning. <ul style="list-style-type: none">● Define the Q-value and explain how it is updated during learning.● Why is the learning rate (α) important in this update process? | 1+2+1 | CO2 |
| | c. What is the significance of Interpretable machine learning? Write some techniques of Interpretability. What is the basic distinction between Grid search and random search in case of hyper-parameter tuning? | 2+2 | CO4 |

- Q.3**
- a. What is the importance of data annotation in machine learning, and how does it contribute to the performance of models? 2 CO1
- b. Suppose you have three different classifiers (Decision Tree, SVM, and Logistic Regression) for a classification task. How can you combine their predictions using **bagging** and **boosting**? Explain with proper diagram. 2+2 CO2
- c. A hospital develops a machine learning model to detect cancer from patient scans. Out of 1000 patients: 3+1 CO4
- 100 patients actually have cancer.
 - The model predicts 120 patients as having cancer. Among them, 80 actually have cancer.
- i. Calculate the precision and recall of the model.
- ii. Which metric (precision or recall) would you prioritize in a cancer screening system and why?
- Q.4**
- a. Explain the concept of a labeled dataset and its significance in the context of supervised learning algorithms. 2 CO1
- b. Suppose you are building a Transformer-based model for **machine translation** from English to French: 1+2+1 CO2
- Which part of the Transformer processes the English input, and which part generates the French output?
 - Why is positional-encoding necessary in both encoder and decoder?
 - What is the role of Masked Self-Attention in Decoder?
- c. Draw the ROC Curve for the following Table 1: 4 CO4

<i>Tuple #</i>	<i>Class</i>	<i>Prob.</i>
1	<i>P</i>	0.90
2	<i>P</i>	0.80
3	<i>N</i>	0.70
4	<i>P</i>	0.60
5	<i>P</i>	0.55
6	<i>N</i>	0.54
7	<i>N</i>	0.53
8	<i>N</i>	0.51
9	<i>P</i>	0.50
10	<i>N</i>	0.40

Table 1: Information on positive and negative samples

- Q.5**
- a.** How does machine learning enable artificial intelligence systems to adapt and improve their performance over time, and what role does data play in this process? **2** **CO1**
 - b.** What is the key idea of Bidirectional Encoder Representations from Transformers (BERT)? Describe how you would **fine-tune** BERT for a binary sentiment classification task (positive/negative). **4** **CO2**
 - c.** In addition to accuracy-based measures, classifiers can also be compared with respect to the additional aspects. What are the additional aspects? Explain. Why do need to perform the statistical test? **3+1** **CO4**
- Q.6**
- a.** In what scenarios would you recommend using deep learning over machine learning, and vice versa, considering factors like data complexity, computational resources, and interpretability? **2** **CO1**
 - b.** What is the main difference between **PCA** and **LDA** in terms of their purpose in data analysis? Suppose you have a dataset with 3 classes and 5 features. Explain how **PCA** reduces the dimensionality of the dataset. **1+3** **CO2**
 - c.** What is the significance of Confusion Matrix? Consider a confusion matrix: **1+3** **CO4**

Classes	Yes	No
Yes	90	210
No	140	9560

Calculate the following metrics from above confusion matrix:

- i) Specificity
- ii) Sensitivity
- iii) Overall accuracy

Premier University
Department of Computer Science & Engineering
8th Semester Final Exam, October 2025
Session: Spring 2025

Course Code: CSE – 481 Course Title: Contemporary Course of Computer Science
Total Marks: 40 Time: 3 Hours

NB: Answer any of four (4) questions from following six (6) questions.

Each question carries equal marks.

- Q.1** a. Write down the differences between the Traditional Computing Model and the Cloud Computing Model. **4 CO3**
- b. According to the following Figure 1, which EC2 Reserved Instance is most beneficial? Justify your answer by comparing it to others. **3 CO2**

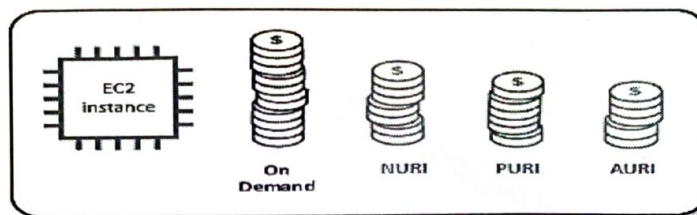


Figure 1: Reserve Instance comparison with on-demand instance.

- c. What is the functionality of Points of Presence (POP) in Amazon CloudFront? What are the components of it? **2+1 CO1**
- Q.2** a. What are Web Services? Draw a simple AWS cloud solution combining Network, Compute, Database, and Storage services as building blocks. **1+3 CO1**
- b. Write down the names of five services for which AWS does not impose any charge for individual use. From those find the services that may be charged when used with other services? **2.5+ CO2**
1.5
- c. Write the names of eight services in AWS, which are used as Compute Services. **2 CO5**
- Q.3** a. Compare among Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) with examples. **3 CO3**
- b. Suppose a VPC with an IPv4 CIDR block of 172.16.0.0/16 has 65,536 total IP addresses. A subnet in this VPC uses an IPv4 CIDR block of 172.16.50.0/24. Now write down the IP addresses and their names that are reserved in this subnet. **2 CO5**
- c. What are the nine key decisions to make when you create an EC2 instance by using the AWS Management Console Launch Instance Wizard? Briefly describe each decision step. **5 CO2**
- Q.4** a. What is the requirement to authenticate as an IAM user to gain access using Programmatic Access and AWS Management Console access? **4 CO4**
- b. Identify the 16 components of cloud from the following Figure 2. **4 CO2**

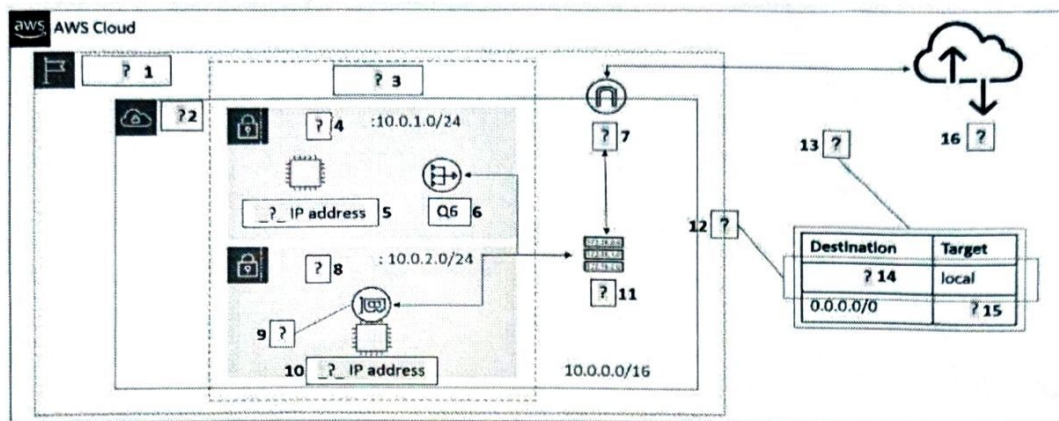


Figure 2: Label this network diagram

c. Draw the Amazon EC2 instance lifecycle. 2 CO5

Q.5 a. Differentiate among Amazon EBS volume types with respect to Maximum Volume Size, Maximum IOPS/Volume, and Maximum Throughput/Volume. Write down the EBS features. 2+2 CO3

b. What are the maintenance challenges of relational databases? 3 CO4

c. Which types of load balancers are needed in the following cases? 3 CO4

- You must support traffic to a containerized application.
- You have extremely spiky and unpredictable TCP traffic.
- You need simple load balancing with multiple protocols.
- You need to support a static or Elastic IP address, or an IP target outside a VPC.
- You need a load balancer that can handle millions of requests per second while maintaining low latencies.
- You must support HTTPS requests.

Q.6 a. A collaboration software company processes email for enterprise customers. They have more than 250 enterprise customers and more than half a million users. They must store petabytes of data for their customers. The following Figure 3 illustrates their architecture. Which storage type can they use? Justify your answer. 3 CO4

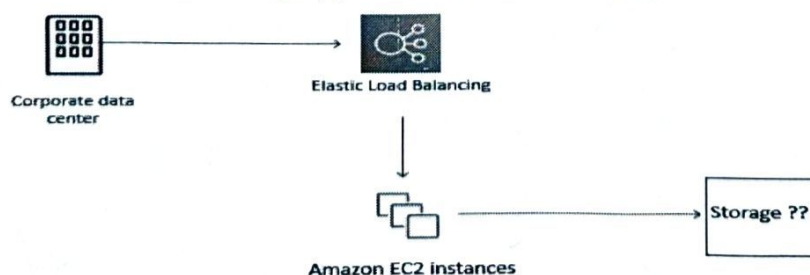


Figure 3: Storage selection scenario of a software company

b. Differentiate between Relational versus non-relational databases. 3 CO3

c. How does Amazon EC2 Auto Scaling work? Explain with a figure. 4 CO1

Premier University
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8th Semester (B.Sc.) Final Examination, October 2025

Course Code: CSE 459

Course Title: Pattern Recognition

Time: 3 Hours

Full Marks: 40

NB: Answer any of four (4) from the following six (6) questions.

Each question carries equal marks.

- Q1. a. Suppose you are designing a smartphone app that unlocks using facial recognition. Which pattern recognition features would be most important, and how? 5 CO2
- b. Analyze the effect of different activation functions on network performance and the advantage of one on other. 5 CO4
- Q2. a. A fruit sorting system uses camera sensors to group fruits by size and color. Identify whether this is supervised or unsupervised classification and justify your answer and also write the difference between them 5 CO3
- b. Consider a perceptron with inputs [2, 3, 4], weights [0.5, -0.6, 0.2], and bias = 1. Compute the weighted sum and final output if the activation function is **ReLU** and **Sigmoid** (show intermediate steps) 5 CO3
- Q3. a. 6 CO4

Day	Outlook	Temperature	Routine	Wear Coat?
D_1	Sunny	Cold	Indoor	No
D_2	Sunny	Warm	Outdoor	No
D_3	Cloudy	Warm	Indoor	No
D_4	Sunny	Warm	Indoor	No
D_5	Cloudy	Cold	Indoor	Yes
D_6	Cloudy	Cold	Outdoor	Yes
D_7	Sunny	Cold	Outdoor	Yes

Find the probability of $X=(\text{Cloudy, Warm, Outdoor})$ and show each step of computation

- b. Write the assumptions in Naïve Bayes Algorithm and how these assumptions helps explain with advantage. 4 CO5

- Q4. a. Demonstrate a generic flowchart for K-Means clustering mechanism. 3 CO1
- b. A logistics company has 4 delivery locations at coordinates $(0,0)$, $(0,1)$, $(5,5)$, and $(5,4)$ in a city grid. The company wants to establish 2 distribution hubs to minimize delivery distances. Using K-Means clustering with $K=2$, determine the optimal hub locations and which delivery points each hub should serve. 7 CO3
- Q5. a. A university campus has identified 6 high-traffic areas at coordinates $(0,0)$, $(0,1)$, $(5,4)$, $(5,5)$, $(4,5)$, and $(1,0)$ where students frequently use WiFi. The network team needs to install WiFi access points, where each access point can effectively cover areas within a 2-unit radius ($T=2$). Using the cluster seeking algorithm, determine how many access points are needed and which areas each will serve. 6 CO3
- b. Describe what makes essentially *Cluster Seeking* algorithm and *Maximin Distance* algorithm partitional clustering. 4 CO1
- Q6. a. Explain the necessity of syntactic pattern recognition in textual data format. 4 CO1
- b. Given the following Finite State Grammar G: 6 CO2
- $$\begin{aligned}
 S &\rightarrow aA \\
 S &\rightarrow bB \\
 A &\rightarrow cB \\
 B &\rightarrow a \\
 A &\rightarrow c
 \end{aligned}$$

Draw the graphical representation of this grammar showing:

- All nonterminal nodes (S, A, B)
- The terminal node (T)
- Directed edges representing productions between non-terminals and terminals